

Alex Curran

24-3-22

Stat 453 HW9

Ch 9: 4, 7, 10, 13, 14, 15, 16

4. $t=4$ $n_1=5$ $n_2=6$ $n_3=4$ $n_4=5$

$$\hat{l}_1 = \bar{y}_1 - 3\bar{y}_2 + \bar{y}_3 + \bar{y}_4$$

$$\hat{l}_2 = \bar{y}_1 + \bar{y}_2 - 2\bar{y}_3$$

$$\hat{l}_3 = \bar{y}_1 + \bar{y}_2 + \bar{y}_3 + \bar{y}_4$$

$$\hat{l}_4 = \bar{y}_1 + \bar{y}_2 - 3\bar{y}_3 + \bar{y}_4$$

a) $\hat{l}_1, \hat{l}_2, \text{ or } \hat{l}_4$ are contrasts (b/c coefficients sum to 0)

b) ~~$\hat{l}_1, \hat{l}_2, \text{ or } \hat{l}_4$ are contrasts~~

$\hat{l}_1, \hat{l}_2, \text{ or } \hat{l}_4$; $\hat{l}_3$; $\hat{l}_2$; $\hat{l}_4$

$$\frac{1(1)}{5} - \frac{3(1)}{6} + \frac{1(1)}{4} - \frac{1(1)}{5} ; \frac{1(1)}{5} - \frac{3(1)}{6} - \frac{3(1)}{4} + \frac{1(1)}{5} ; \frac{1(1)}{5} + \frac{1(1)}{6} - \frac{0(1)}{4} - \frac{2(1)}{5}$$

$\neq 0$, not orthogonal ; $\neq 0$, not orthogonal ; $\neq 0$, not orthogonal

none orthogonal

7. Example 8.6 NE $\rightarrow \bar{y}_1$ SE $\rightarrow \bar{y}_2$ NW $\rightarrow \bar{y}_3$ W $\rightarrow \bar{y}_4$

a) $l = \frac{1}{2}\bar{y}_1 + \frac{1}{2}\bar{y}_2 - \bar{y}_4$ east compare to west

or this includes NW for which bc $l = \frac{1}{2}\bar{y}_1 + \frac{1}{2}\bar{y}_2 - \frac{1}{2}\bar{y}_3 - \frac{1}{2}\bar{y}_4$

b) $l = \bar{y}_1 - \frac{1}{3}\bar{y}_2 - \frac{1}{3}\bar{y}_3 - \frac{1}{3}\bar{y}_4$ or $3\bar{y}_1 - \bar{y}_2 - \bar{y}_3 - \bar{y}_4$ NE compare to rest

c) $l = \bar{y}_1 - \bar{y}_2$ NE compare to SE

d) $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ vs $H_1: \text{ALDI}$ $\alpha = 0.05$ $\bar{y}_1 = .827$ $\bar{y}_2 = .343$ $\bar{y}_3 = .985$ $\bar{y}_4 = .120$
 $n = 6$ for all groups overall mean = .46875

$$SSC = \frac{\sum \hat{l}_i^2}{\sum \frac{1}{n_i}}$$

Contrast	a_1	a_2	a_3	a_4	$\sum a_i^2$	$\hat{l}$	SSC	SSC/SSB	SSC/MSE
1 East vs West	1	1	-1	-1	4	.465	.3243	.1935	1.93035
2 NE vs SW	3	-1	-1	1	12	1.433	1.0267	.6126	6.1131
3 NE vs SE	1	-1	0	0	2	.484	.7027	.4143	2.4159

probably more a mistake

$$SSB = 6(.827 - .46875)^2 + 6(.343 - .46875)^2 + 6(.985 - .46875)^2 + 6(.12 - .46875)^2 = 1.67776$$

Critical Value: $F_{1,20,0.05} = 4.35$

$$MSE = MSW = \frac{SSW}{21} = .00168$$

Contrast	Conclusion
E vs W	reject null
NE vs SW	reject null
NE vs SE	reject null

e) 1 3 2 : 3 -1 4 1 +1 $\neq 0$, not orthogonal

2 3 3 : 3 +1 $\neq 0$, not orthogonal

1 3 3 : 1 -1 = 0, orthogonal

Contrasts not mutually orthogonal, but East vs West vs NE vs SE are orthogonal

10. in 9.8, $\alpha_E \neq 0.05$

- The Bonferroni procedure is very conservative, making it harder to reject null hypotheses (could lead to high rate of false negatives (Type II errors))
- Becomes even more conservative when more tests are conducted, since the number of tests m is used to divide the original Type I error rate α
- Bonferroni procedure depends upon number of other tests run, which affects the interpretation of findings
- Procedure is consistent w/ general null hypothesis that all nulls (for each test) are true, but often unrealistic or unhelpful in real scenarios

13. Run ANOVA to determine if there were any significant differences among five weight-reducing agents.

Use $\alpha = 0.05$ $n = 10$ $N = 50$

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ H_a : At least one inequality

Total sample mean: 10.97

Total sample variance: 2.1996

$$\bar{A}_1 = 12.05$$

$$\bar{A}_2 = 11.02$$

$$\bar{A}_3 = 10.27$$

$$\bar{A}_4 = 12.24$$

$$\bar{S} = 9.27$$

$$S_1^2 = 0.6872$$

$$S_2^2 = 1.2573$$

$$S_3^2 = 1.0534$$

$$S_4^2 = 0.5715$$

$$S_5^2 = 1.3423$$

$$SSB = \sum_{i=1}^5 n_i (\bar{z}_i - \bar{z}_{..})^2 = 10(12.05 - 10.97)^2 + 10(11.02 - 10.97)^2 + 10(10.27 - 10.97)^2 + 10(12.24 - 10.97)^2 + 10(9.27 - 10.97)^2$$

$$= 61.618$$

$$SSW = \sum_{i=1}^5 (n_i - 1) S_i^2 = 9(0.6872) + 9(1.2573) + 9(1.0534) + 9(0.5715) + 9(1.3423)$$

$$= 44.2053$$

$$MSB = \frac{SSB}{f-1} = \frac{61.618}{4} = 15.4045$$

$$MSW = \frac{SSW}{N-f} = \frac{44.2053}{50-5} = 0.98234$$

$$F = \frac{MSB}{MSW} = 15.6814$$

$$\text{Critical value: } F_{\alpha, 4, 45} = 2.579$$

$F > CV$, so there is sufficient evidence to reject the null hypothesis and conclude that at least one mean value is different from the others

Only assumption that appears to be violated is the assumption that all samples have a similar variance. The sample variances were not relatively close together, which could be a violation of the ANOVA assumptions

Conclusions: At least one of the weight loss agents is more effective than one or all of the others. From raw data, all $A_1, \dots, A_4$ seem to be more effective than S

which contrasts are
significant?

$$SSC = \frac{\hat{\ell}^2}{\sum \frac{a_i^2}{n_i}}$$

13. Linear contrasts (set up for #15)

mean pairs:

A_1 vs A_2 A_2 vs A_3 A_3 vs A_4 A_4 vs S
 A_1 vs A_3 A_2 vs A_4 A_3 vs S
 A_1 vs A_4 A_2 vs S
 A_1 vs S

10 total pairs

trsr	CONTRAST	a_1	a_2	a_3	a_4	a_5	$\sum a_i^2$	$\hat{\ell}$	SSC	SSC/SSB	SSC/MSE
1	A_1 vs A_2	1	-1	0	0	0	2	1.03			
2	A_1 vs A_3	1	0	-1	0	0	2	1.78			
3	A_1 vs A_4	1	0	0	-1	0	2	-0.19			
4	A_1 vs S	1	0	0	0	-1	2	2.78			
5	A_2 vs A_3	0	1	-1	0	0	2	0.75			
6	A_2 vs A_4	0	1	0	-1	0	2	-1.22			
7	A_2 vs S	0	1	0	0	-1	2	1.75			
8	A_3 vs A_4	0	0	1	-1	0	2	-1.97			
9	A_3 vs S	0	0	1	0	-1	2	1			
10	A_4 vs S	0	0	0	1	-1	2	2.97			

9.14. use Tukey's w w/ $\alpha = 0.05$ to determine significant different pairs of means

ordered means: $\bar{A}_1 = 12.24$

$\bar{A}_2 = 12.05$

$\bar{A}_3 = 11.02$

$\bar{A}_4 = 10.27$

$\bar{S} = 9.27$

Regression line
for μ_1

Comparison table $(\mu_1 - \mu_2)$ compare to w
treatment

$\mu_1 \backslash \mu_2$	A_1	A_2	A_3	A_4	A_5
A_1	N/A	N/A	N/A	Reject	Reject
A_2	N/A	N/A	N/A	Reject	Reject
A_3	N/A	N/A	N/A	N/A	Reject
A_4	Reject	Reject	N/A	N/A	N/A
S	Reject	Reject	Reject	N/A	N/A

$$TS = \frac{\bar{J}_{largest} - \bar{J}_{smallest}}{S_p \sqrt{1/n}}$$

$n=6, S_p =$

$$w = q_{\alpha, E(t, df)} \sqrt{\frac{S_w^2}{n}}$$

$$= 11.04$$

$$= 1.63469$$

$$S_w^2 = MSW = .98234$$

$n=6$

$$t = 5, df = 24, \alpha_E = 0.05$$

Conclusions: All groups are significantly different to the S group aside from group 4. Group 1 is also significantly different to group 4. Group 2 is significantly different to group 4 as well.

15. a) use binomial test, $\alpha_B = 0.05$ — α_B under $\alpha = 0.05$

$$S_p^2 = \frac{\sum (n_i - 1) S_i^2}{\sum (n_i - 1)} = \frac{9(0.6672) + 9(1.2773) + 9(1.0734) + 9(0.5769) + 9(1.3423)}{40} = 1.1041$$

$$TS: \frac{|\bar{J}_i - \bar{J}_j|}{1.0512 \sqrt{\frac{1}{n}}} = \frac{|\bar{J}_i - \bar{J}_j|}{0.47013} > 1.664$$

Comparison	TS	Reject?
A_1 vs A_2	2.14	X
A_1 vs A_3	3.78	X
A_1 vs A_4	6.404	X

Comparison	TS	Reject?
A_2 vs A_3	1.883	X
A_2 vs A_4	2.59	X

Comparison	TS	Reject?
A_3 vs A_4	3.7	X
A_3 vs S	4.14	X
A_4 vs S	6.32	X

b) use Scheffe's S procedure $\alpha_E = 0.05$

$$S = \sqrt{1/n} \sqrt{15 - 11F_{0.05, 4, 24}}$$

$$F\text{-value: } 2.979$$

$$w = \sqrt{S^2 \left(\frac{q_{\alpha, E(t, df)}^2}{n} \right)}$$

Variables

Contrast	1	2	3	4	5	6	7	8	9	10
$\hat{\mu}$	1.03	1.7	-0.19	2.76	0.75	-2.2	1.75	1.97	1	-2.97
$V(\hat{\mu})$				1.44						
$V(\hat{\mu})$					1.4236					
Decision	X	X	X	X	X	X	X	X	X	X

$\mu = \mu_1$ null hypothesis $\mu_1 = \mu_2$, $\mu \neq \mu_2$ reject null hypothesis $\mu_1 \neq \mu_2$

c) Which of the tests determined largest to or significantly different pairs? Which had the fewest?

All of the tests I conducted resulted in a null hypothesis being rejected. Bonferroni's test had the fewest significant different pairs, and Scheffe's the most.

overarching formulas
for this problem:

$$TS = \frac{\mu_{A_i} - 0.27}{\sqrt{\frac{s_{A_i}^2}{10} + \frac{1.3423}{10}}}$$

assuming $\sigma_{A_i}^2 \neq \sigma_S^2$

df = $81 / (1/(n-1)^2 + 1/n^2)$ where $C = \frac{s_{A_i}^2 / 10}{s_{A_i}^2 / 10 + 1.3423 / 10}$

These used in all calculations below; df and C noted down

16. use $\alpha = 0.05$ to determine which of the new agents produced a significantly larger mean weight loss in comparison to the standard agent

Test 1: A_1 vs S	Test 2: A_2 vs S	Test 3: A_3 vs S	Test 4: A_4 vs S
$H_0: \mu_{A_1} = \mu_S$ vs $H_a: \mu_{A_1} \neq \mu_S$	$H_0: \mu_{A_2} = \mu_S$ vs $H_a: \mu_{A_2} \neq \mu_S$	$H_0: \mu_{A_3} = \mu_S$ vs $H_a: \mu_{A_3} \neq \mu_S$	$H_0: \mu_{A_4} = \mu_S$ vs $H_a: \mu_{A_4} \neq \mu_S$
TS = 6.1704	TS = 3.4323	TS = 2.04	TS = 6.74
$CV = t_{16, 0.025} = 2.12$ TS > CV	$CV = t_{17, 0.025} = 2.11$ TS > CV	$CV = 2.11 = t_{17, 0.025}$ TS < CV	$CV = t_{99, 0.025} = 2.131$ TS > CV
Decision: Reject H_0	Decision: Reject H_0	Decision: Do <u>not</u> reject H_0	Decision: Reject H_0

Conclusions: Agents 1, 2, and 4 are significantly different in comparison to the standard agent